

0.1 Riemann-Stieltjes Integrability Criteria and Measure-Zero Discontinuities

We generalize the classical Riemann integral by integrating a function f with respect to a monotonically increasing function α . This Riemann-Stieltjes framework provides the theoretical foundation for probability theory and unified functional analysis.

Let $[a, b]$ be a closed interval. A **partition** P of $[a, b]$ is a finite set of points $P = \{x_0, x_1, \dots, x_n\}$ such that $a = x_0 < x_1 < \dots < x_n = b$. We define $\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1})$. For a bounded function $f : [a, b] \rightarrow \mathbb{R}$, we define the upper and lower Darboux-Stieltjes sums with respect to P and α as:

$$U(P, f, \alpha) = \sum_{i=1}^n M_i \Delta\alpha_i, \quad L(P, f, \alpha) = \sum_{i=1}^n m_i \Delta\alpha_i$$

where $M_i = \sup_{x \in [x_{i-1}, x_i]} f(x)$ and $m_i = \inf_{x \in [x_{i-1}, x_i]} f(x)$.

We define the upper and lower integrals as the infimum and supremum of these sums over all possible partitions, respectively. If these two values coincide, we say f is Riemann-Stieltjes integrable with respect to α , denoted $f \in \mathcal{R}(\alpha)$.

[Cauchy Criterion for Integrability] Let f be bounded on $[a, b]$, and let α be monotonically increasing. Then $f \in \mathcal{R}(\alpha)$ if and only if for every $\epsilon > 0$, there exists a partition P such that:

$$U(P, f, \alpha) - L(P, f, \alpha) < \epsilon$$

() Assume that for every $\epsilon > 0$, there exists a partition P satisfying the inequality. By definition, the upper integral $\overline{\int} f d\alpha$ and lower integral $\underline{\int} f d\alpha$ satisfy:

$$L(P, f, \alpha) \leq \underline{\int} f d\alpha \leq \overline{\int} f d\alpha \leq U(P, f, \alpha)$$

Thus, we have:

$$0 \leq \overline{\int} f d\alpha - \underline{\int} f d\alpha \leq U(P, f, \alpha) - L(P, f, \alpha) < \epsilon$$

Since this holds for all $\epsilon > 0$, the upper and lower integrals must be equal, proving $f \in \mathcal{R}(\alpha)$.

() Assume $f \in \mathcal{R}(\alpha)$, and let $I = \int f d\alpha$. Let $\epsilon > 0$. By the definitions of the supremum and infimum, there exist partitions P_1 and P_2 such that:

$$U(P_1, f, \alpha) < I + \frac{\epsilon}{2} \quad \text{and} \quad L(P_2, f, \alpha) > I - \frac{\epsilon}{2}$$

Let $P = P_1 \cup P_2$ be the common refinement. Because refining a partition can only decrease upper sums and increase lower sums, we have:

$$U(P, f, \alpha) \leq U(P_1, f, \alpha) < I + \frac{\epsilon}{2}$$

$$L(P, f, \alpha) \geq L(P_2, f, \alpha) > I - \frac{\epsilon}{2}$$

Subtracting the lower sum inequality from the upper sum inequality yields:

$$U(P, f, \alpha) - L(P, f, \alpha) < \left(I + \frac{\epsilon}{2}\right) - \left(I - \frac{\epsilon}{2}\right) = \epsilon$$

To characterize precisely which functions are integrable, we must analyze their points of discontinuity. For the standard Riemann integral where $\alpha(x) = x$, Lebesgue provided an exact characterization using the concept of measure zero.

A set $E \subset \mathbb{R}$ has **Lebesgue measure zero** if for every $\epsilon > 0$, there exists a countable collection of open intervals $\{(a_k, b_k)\}_{k=1}^{\infty}$ such that $E \subset \bigcup_{k=1}^{\infty} (a_k, b_k)$ and $\sum_{k=1}^{\infty} (b_k - a_k) < \epsilon$.

The **oscillation** of a bounded function f on an interval J is $\omega_f(J) = \sup_J f - \inf_J f$. The oscillation of f at a point x is defined as $\omega_f(x) = \lim_{\delta \rightarrow 0} \omega_f((x - \delta, x + \delta))$.

It is a standard topological result that f is continuous at x if and only if $\omega_f(x) = 0$. Furthermore, for any $\eta > 0$, the set of points $D_\eta = \{x \in [a, b] \mid \omega_f(x) \geq \eta\}$ is closed.

[Lebesgue's Criterion - Sufficiency] Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. If the set D of discontinuities of f has Lebesgue measure zero, then f is Riemann integrable on $[a, b]$.

Let $M = \sup_{x \in [a, b]} |f(x)|$. If $M = 0$, f is identically zero and trivially integrable. Assume $M > 0$. Let $\epsilon > 0$ be given. We must construct a partition P such that $U(P, f) - L(P, f) < \epsilon$.

Let $\eta = \frac{\epsilon}{2(b-a)}$. Consider the set $D_\eta = \{x \in [a, b] \mid \omega_f(x) \geq \eta\}$. Because D_η is a closed subset of the compact interval $[a, b]$, D_η is compact. Furthermore, since $D_\eta \subset D$ and D has measure zero, D_η also has measure zero.

Therefore, there exists a countable collection of open intervals covering D_η with total length strictly less than $\frac{\epsilon}{4M}$. Because D_η is compact, we can extract a finite subcover $\mathcal{V} = \{V_1, \dots, V_k\}$. Let $V = \bigcup_{i=1}^k V_i$. The sum of the lengths of the intervals in $\mathcal{V}$ is also less than $\frac{\epsilon}{4M}$.

Now consider the set $K = [a, b] \setminus V$. Since V is open, K is closed and bounded, thus compact. For every $x \in K$, we have $x \notin D_\eta$, which means $\omega_f(x) < \eta$. By the definition of oscillation at a point, there exists an open interval U_x containing x such that $\omega_f(U_x) < \eta$. The collection $\{U_x\}_{x \in K}$ forms an open cover of the compact set K . Thus, there exists a finite subcover $\mathcal{U} = \{U_{x_1}, \dots, U_{x_m}\}$.

The endpoints of the intervals in the finite covers $\mathcal{V}$ and $\mathcal{U}$ that intersect $[a, b]$ define a partition $P = \{t_0, t_1, \dots, t_N\}$ of $[a, b]$. We separate the subintervals $[t_{j-1}, t_j]$ into two disjoint classes:

1. Class $\mathcal{A}$ contains those subintervals that intersect D_η . By construction, these are entirely contained within V .
2. Class $\mathcal{B}$ contains all remaining subintervals. These do not intersect D_η , and are entirely contained within some interval from $\mathcal{U}$.

We now bound the difference between the upper and lower sums:

$$U(P, f) - L(P, f) = \sum_{j=1}^N (M_j - m_j) \Delta t_j = \sum_{j \in \mathcal{A}} (M_j - m_j) \Delta t_j + \sum_{j \in \mathcal{B}} (M_j - m_j) \Delta t_j$$

For subintervals in Class $\mathcal{A}$, the maximum difference $M_j - m_j$ is bounded trivially by $2M$. The sum of their lengths Δt_j is bounded by the total length of $\mathcal{V}$, which is less than $\frac{\epsilon}{4M}$. Thus:

$$\sum_{j \in \mathcal{A}} (M_j - m_j) \Delta t_j \leq 2M \sum_{j \in \mathcal{A}} \Delta t_j < 2M \left(\frac{\epsilon}{4M} \right) = \frac{\epsilon}{2}$$

For subintervals in Class $\mathcal{B}$, each is contained in an interval where the oscillation is strictly less than η . Hence, $M_j - m_j < \eta$. The sum of their lengths is bounded by the total length of $[a, b]$, which is $b - a$. Thus:

$$\sum_{j \in \mathcal{B}} (M_j - m_j) \Delta t_j < \eta \sum_{j \in \mathcal{B}} \Delta t_j \leq \eta(b - a) = \frac{\epsilon}{2(b - a)}(b - a) = \frac{\epsilon}{2}$$

Adding the contributions from both classes yields:

$$U(P, f) - L(P, f) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

By the Cauchy criterion, f is Riemann integrable on $[a, b]$.